ROHIT K

Email: rohitvanjre@gmail.com
Phone: +91 8660670557
LinkedIn: linkedin.com/in/rohit-k-vanjre-742432372
GitHub: github.com/rohitkvanjre
Portfolio: rohit-k-vanjre-portfolio.vercel.app
Location: Anekal, Bengaluru

Summary

Computer Science and Engineering B.E. (2026) with hands-on experience in Python application development, computer-vision attendance automation, Windows systems software and Android app development (Java, XML, Room). Built a Windows software deployment system with TLS-encrypted communication and device pairing and co-authored a published research paper on automated lab software deployment (IJIRIS, 2025).

Education

Bachelor of Engineering (B.E) in Computer Science and Engineering **2023 – 2026**
Sri Sairam College of Engineering, Anekal, Bengaluru CGPA: 7.31/10

Diploma in Computer Science and Engineering **2020 – 2023**
Sri Jayachamarajendra (Govt.) Polytechnic, Bengaluru CGPA: 8.67/10

Experience

Android App Development Intern (Gen AI, Remote) | MindMatrix, Bengaluru **Feb 2026 – Jul 2026**

- Developed Android application screens in Java using XML layouts, fragments, Room, ViewModel and LiveData.
- Used generative AI tools to accelerate coding, debugging and documentation.

Intern | Analogica Software Development Pvt. Ltd. **Jun 2023 – Aug 2023**

- Worked on DBMS, Python and data structures; applied basic Statistics, AI and ML concepts.

Projects

1. **Smart Network-Based Automated Software Deployment for Multi-System Lab Environments | GitHub |**

   Technologies used: Python, Multithreading, Socket Programming (TCP/UDP), FTP, TLS/SSL, SQLite, Tkinter, Windows Services (pywin32), psutil, PyInstaller, Inno Setup

   - Built a distributed client-server (master–agent) system that automates silent software deployment across lab PCs, with 15+ remote management operations.
   - Auto-detects 5 installer types (MSI, NSIS, Inno Setup, InstallShield, WiX) and checks every file with a SHA-256 hash before install.
   - Secured the system with TLS encryption, certificate pinning and device authentication: pairing codes with a 30-minute expiry and rate limiting, plus a revocable token for each PC.
   - Free, open-source alternative to commercial endpoint-management tools that cost an estimated $1,500–5,000+ per year for 50 systems

2. **Intelligent Attendance Automation with AI | GitHub |**

   Technologies used: Python, Flask, Jinja2, SQLite, OpenCV, InsightFace (ArcFace), MTCNN, PyTorch, ONNX Runtime, pandas

   - Built a live face recognition pipeline using OpenCV and InsightFace: ArcFace 512-d embeddings from 30 enrolment photos per student (3 poses × 10), matched from a webcam by cosine similarity at a 0.5 threshold.
   - Implemented attendance logic: 6 recognition passes per class session, Present if matched in at least 4, saved to SQLite and exported as an Excel report per session.
   - Developed a Flask web app (26 routes, 16 Jinja2 templates) for student management, webcam capture, timetable editing, and attendance viewing and download.
   - Implemented role-based access control with 6 scoped roles (admin, HoD, class coordinator, subject teacher, mentor, student management), scrypt password hashing, and permission-checked downloads.
   - Designed a 10-table relational schema (departments, classes, subjects, timetable, users and roles, attendance) and added Windows and WSL2 support with USB-webcam passthrough and one-click launch scripts.

Publication and Technical Writing

- Rohit K. **"**Unveiling the Magic of Matplotlib: Exploring Python's Data Visualization Library**"** Certisured – June 2023. https://certisured.com/blogs/matplotlib
- Karthika K, Rohit K et al. "Smart Network-Based Automated Software Deployment for Multi-System Lab Environments", IJIRIS Journal, Vol. 11, Issue 08, Nov 2025. https://doi.org/10.26562/ijiris.2025.v1108.25

Certifications

- Pandas and NumPy Tips, Tricks, and Techniques — Infosys Springboard, 2024
- Data Visualization in Python by Examples — Infosys Springboard, 2024
- Introduction to Data Science — Infosys Springboard, 2024

Skills

- **Languages and Web:** Python, Java, C, Flask, Jinja2, HTML, CSS, JavaScript (basic)
- **Android Development:** Android Studio, Java, XML layouts, Room, ViewModel, LiveData
- **Databases and Libraries:** SQLite, OpenCV, InsightFace, Pandas, NumPy, Matplotlib
- **Networking and Security:** Socket Programming (TCP/UDP), TLS, FTP, SHA-256 File Verification, Silent Software Installation
- **Tools and Concepts:** SDLC, PyInstaller, Inno Setup, pywin32, AI-assisted development, Data Structures, Algorithms
- **Familiar With:** MySQL, Seaborn, Figma